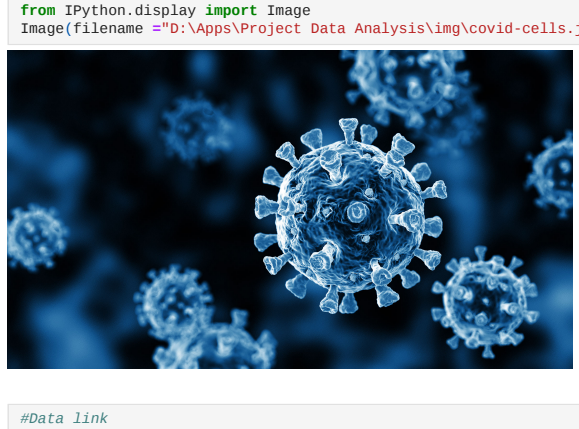
COVID - 19 (Analysis,Predicting)

In [2]:
```python
#Display a picture
from IPython.display import Image
Image(filename ="D:\Apps\Project Data Analysis\img\covid-cells.jpg",width=500,height=500)
```

Out[2]:



In [3]:
```python
#Data link
'''https://www.kaggle.com/imdevskp/corona-virus-report'''
```

Out[3]: 'https://www.kaggle.com/imdevskp/corona-virus-report'

In [4]:
```python
#install libraries
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
%matplotlib inline


#Lets install the data in order read it
#our data has four necessary files
data = pd.read_csv(r"D:\Apps\Data_Analysis\Data_Covid\country_wise_latest.csv")
data2 = pd.read_csv(r"D:\Apps\Data_Analysis\Data_Covid\covid_19_clean_complete.csv")
data3 = pd.read_csv(r"D:\Apps\Data_Analysis\Data_Covid\day_wise.csv")
data4 = pd.read_csv(r"D:\Apps\Data_Analysis\Data_Covid\worldometer_data.csv")
```

country_wise_lates

In [5]:
```python
#data 1 = country_wise_late
#Firstly replace column name
data.columns = data.columns.str.replace(' ','_')
data.columns = data.columns.str.replace(' ','_')
#replace the nan value
data = data.replace('', np.nan).fillna(0)
data.info()
data.head(10)
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 187 entries, 0 to 186
Data columns (total 15 columns):
 #   Column                  Non-Null Count  Dtype  
---  ------                  --------------  -----  
 0   Country/Region          187 non-null    object 
 1   Confirmed               187 non-null    int64  
 2   Deaths                  187 non-null    int64  
 3   Recovered               187 non-null    int64  
 4   Active                  187 non-null    int64  
 5   New_cases               187 non-null    int64  
 6   New_deaths              187 non-null    int64  
 7   New_recovered           187 non-null    int64  
 8   Deaths_/_100_Cases      187 non-null    float64
 9   Recovered_/_100_Cases   187 non-null    float64
 10  Deaths_/_100_Recovered  187 non-null    float64
 11  Confirmed_last_week     187 non-null    int64  
 12  1_week_change           187 non-null    int64  
 13  1_week_%_increase       187 non-null    float64
 14  WHO_Region              187 non-null    object 
dtypes: float64(4), int64(9), object(2)
memory usage: 22.0+ KB
```

Out[5]:

	Country/Region	Confirmed	Deaths	Recovered	Active	New_cases	New_deaths	New_recovered	Deaths_/_100_Cases	Recovered_/_100_Cases	Deaths_/_100_Recovered
0	Afghanistan	36263	1269	25198	9796	106	10	18	3.50	69.49	5.04
1	Albania	4880	144	2745	1991	117	6	63	2.95	56.25	5.25
2	Algeria	27973	1163	18837	7973	616	8	749	4.16	67.34	6.17
3	Andorra	907	52	803	52	10	0	0	5.73	88.53	6.48
4	Angola	950	41	242	667	18	1	0	4.32	25.47	16.94
5	Antigua and Barbuda	86	3	65	18	4	0	5	3.49	75.58	4.62
6	Argentina	167416	3059	72575	91782	4890	120	2057	1.83	43.35	4.21
7	Armenia	37390	711	26665	10014	73	6	187	1.90	71.32	2.67
8	Australia	15303	167	9311	5825	368	6	137	1.09	60.84	1.79
9	Austria	20558	713	18246	1599	86	1	37	3.47	88.75	3.91

covid_19_clean_complete

In [6]:
```python
#data 2 = covid_19_clean_complete
data2.columns = data2.columns.str.replace(' ','_')
data2 = data2.replace('', np.nan).fillna(0)
data2.info()
data2.head(10)
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 49068 entries, 0 to 49067
Data columns (total 10 columns):
 #   Column          Non-Null Count  Dtype  
---  ------          --------------  -----  
 0   Province/State  49068 non-null  object 
 1   Country/Region  49068 non-null  object 
 2   Lat             49068 non-null  float64
 3   Long            49068 non-null  float64
 4   Date            49068 non-null  object 
 5   Confirmed       49068 non-null  int64  
 6   Deaths          49068 non-null  int64  
 7   Recovered       49068 non-null  int64  
 8   Active          49068 non-null  int64  
 9   WHO_Region      49068 non-null  object 
dtypes: float64(2), int64(4), object(4)
memory usage: 3.7+ MB
```

Out[6]:

	Province/State	Country/Region	Lat	Long	Date	Confirmed	Deaths	Recovered	Active	WHO_Region
0	0	Afghanistan	33.93911	67.709953	2020-01-22	0	0	0	0	Eastern Mediterranean
1	0	Albania	41.15330	20.168300	2020-01-22	0	0	0	0	Europe
2	0	Algeria	28.03390	1.659600	2020-01-22	0	0	0	0	Africa
3	0	Andorra	42.50630	1.521800	2020-01-22	0	0	0	0	Europe
4	0	Angola	-11.20270	17.873900	2020-01-22	0	0	0	0	Africa
5	0	Antigua and Barbuda	17.06080	-61.796400	2020-01-22	0	0	0	0	Americas
6	0	Argentina	-38.41610	-63.616700	2020-01-22	0	0	0	0	Americas
7	0	Armenia	40.06910	45.038200	2020-01-22	0	0	0	0	Europe
8	Australian Capital Territory	Australia	-35.47350	149.012400	2020-01-22	0	0	0	0	Western Pacific
9	New South Wales	Australia	-33.86880	151.209300	2020-01-22	0	0	0	0	Western Pacific

day_wise

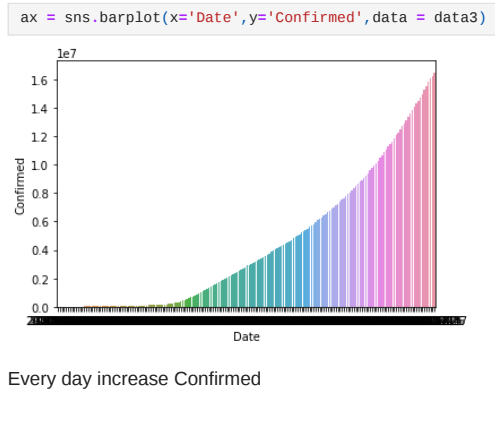
In [7]:
```python
data3.columns = data3.columns.str.replace(' ','_')
data3 = data3.replace('', np.nan).fillna(0)
data3.info()
data3.head(10)
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 188 entries, 0 to 187
Data columns (total 12 columns):
 #   Column                  Non-Null Count  Dtype  
---  ------                  --------------  -----  
 0   Date                    188 non-null    object 
 1   Confirmed               188 non-null    int64  
 2   Deaths                  188 non-null    int64  
 3   Recovered               188 non-null    int64  
 4   Active                  188 non-null    int64  
 5   New_cases               188 non-null    int64  
 6   New_deaths              188 non-null    int64  
 7   New_recovered           188 non-null    int64  
 8   Deaths_/_100_Cases      188 non-null    float64
 9   Recovered_/_100_Cases   188 non-null    float64
 10  Deaths_/_100_Recovered  188 non-null    float64
 11  No._of_countries        188 non-null    int64  
dtypes: float64(3), int64(8), object(1)
memory usage: 17.8+ KB
```

Out[7]:

	Date	Confirmed	Deaths	Recovered	Active	New_cases	New_deaths	New_recovered	Deaths_/_100_Cases	Recovered_/_100_Cases	Deaths_/_100_Recovered	No._of_countries
0	2020-01-22	555	17	28	510	0	0	0	3.06	5.05	60.71	6
1	2020-01-23	654	18	30	606	99	1	2	2.75	4.59	60.00	8
2	2020-01-24	941	26	36	879	287	8	6	2.76	3.83	72.22	9
3	2020-01-25	1434	42	39	1353	493	16	3	2.93	2.72	107.69	11
4	2020-01-26	2118	56	52	2010	684	14	13	2.64	2.46	107.69	13
5	2020-01-27	2927	82	61	2784	809	26	9	2.80	2.08	134.43	16
6	2020-01-28	5578	131	107	5340	2651	49	46	2.35	1.92	122.43	16
7	2020-01-29	6166	133	125	5908	588	2	18	2.16	2.03	106.40	18
8	2020-01-30	8234	171	141	7922	2068	38	16	2.08	1.71	121.28	20
9	2020-01-31	9927	213	219	9495	1693	42	78	2.15	2.21	97.26	24

In [8]:
```python
ax = sns.barplot(x='Date',y='Confirmed',data = data3)
```



Every day increase Confirmed

worldometer_data

In [9]:
```python
data4.columns = data4.columns.str.replace(' ','_')
data4 = data4.replace('', np.nan).fillna(0)
data4.info()
data4.head(10)
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 209 entries, 0 to 208
Data columns (total 16 columns):
 #   Column            Non-Null Count  Dtype  
---  ------            --------------  -----  
 0   Country/Region    209 non-null    object 
 1   Continent         209 non-null    object 
 2   Population        209 non-null    float64
 3   TotalCases        209 non-null    int64  
 4   NewCases          209 non-null    float64
 5   TotalDeaths       209 non-null    float64
 6   NewDeaths         209 non-null    float64
 7   TotalRecovered    209 non-null    float64
 8   NewRecovered      209 non-null    float64
 9   ActiveCases       209 non-null    float64
 10  Serious,Critical  209 non-null    float64
 11  Tot_Cases/1M_pop  209 non-null    float64
 12  Deaths/1M_pop     209 non-null    float64
 13  TotalTests        209 non-null    float64
 14  Tests/1M_pop      209 non-null    float64
 15  WHO_Region        209 non-null    object 
dtypes: float64(12), int64(1), object(3)
memory usage: 26.2+ KB
```

Out[9]:

	Country/Region	Continent	Population	TotalCases	NewCases	TotalDeaths	NewDeaths	TotalRecovered	NewRecovered	ActiveCases	Serious,Critical	Tot_Cases/1M_pop
0	USA	North America	3.311981e+08	5032179	0.0	162804.0	0.0	2576668.0	0.0	2292707.0	18296.0	15194.0
1	Brazil	South America	2.127107e+08	2917562	0.0	98644.0	0.0	2047660.0	0.0	771258.0	8318.0	13716.0
2	India	Asia	1.381345e+09	2025409	0.0	41638.0	0.0	1377384.0	0.0	606387.0	8944.0	1466.0
3	Russia	Europe	1.459409e+08	871894	0.0	14606.0	0.0	676357.0	0.0	180931.0	2300.0	5974.0
4	South Africa	Africa	5.938157e+07	538184	0.0	9604.0	0.0	387316.0	0.0	141264.0	539.0	9063.0
5	Mexico	North America	1.290662e+08	462690	6590.0	50517.0	819.0	308848.0	4140.0	103325.0	3987.0	3585.0
6	Peru	South America	3.301632e+07	455409	0.0	20424.0	0.0	310337.0	0.0	124648.0	1426.0	13793.0
7	Chile	South America	1.913251e+07	366671	0.0	9889.0	0.0	340168.0	0.0	16614.0	1358.0	19165.0
8	Colombia	South America	5.093626e+07	357710	0.0	11939.0	0.0	192355.0	0.0	153416.0	1493.0	7023.0
9	Spain	Europe	4.675665e+07	354530	0.0	28500.0	0.0	0.0	0.0	0.0	617.0	7582.0

We must also check if the information is filled , at least we try with one file

In [10]:
```python
#Columns and row
data.shape
```

Out[10]: (187, 15)

In [11]:
```python
#this function if cell are fill or no
# or use data.isnull().sum()
data.isnull()
```

Out[11]:

	Country/Region	Confirmed	Deaths	Recovered	Active	New_cases	New_deaths	New_recovered	Deaths_/_100_Cases	Recovered_/_100_Cases	Deaths_/_100_Recovered
0	False	False	False	False	False	False	False	False	False	False	False
1	False	False	False	False	False	False	False	False	False	False	False
2	False	False	False	False	False	False	False	False	False	False	False
3	False	False	False	False	False	False	False	False	False	False	False
4	False	False	False	False	False	False	False	False	False	False	False
...	...	...	...	...	...	...	...	...	...	...	...
182	False	False	False	False	False	False	False	False	False	False	False
183	False	False	False	False	False	False	False	False	False	False	False
184	False	False	False	False	False	False	False	False	False	False	False
185	False	False	False	False	False	False	False	False	False	False	False
186	False	False	False	False	False	False	False	False	False	False	False

187 rows × 15 columns

that mean all data fill

In [12]:
```python
#if there are duplicate cells
# or use data.duplicated()
data.duplicated().sum()
```

Out[12]: 0

In [13]:
```python
#Statistical analysis
data.describe()
```

Out[13]:

	Confirmed	Deaths	Recovered	Active	New_cases	New_deaths	New_recovered	Deaths_/_100_Cases	Recovered_/_100_Cases	Deaths_/_100_Recovered
count	1.870000e+02	187.000000	1.870000e+02	1.870000e+02	187.000000	187.000000	187.000000	187.000000	187.000000	187.00
mean	8.813094e+04	3497.518717	5.063148e+04	3.400194e+04	1222.957219	28.957219	933.812834	3.019519	64.820535	inf
std	3.833187e+05	14100.002482	1.901882e+05	2.133262e+05	5710.374790	120.037173	4197.719635	3.454302	26.287694	NaN
min	1.000000e+01	0.000000	0.000000e+00	0.000000e+00	0.000000	0.000000	0.000000	0.000000	0.000000	0.00
25%	1.114000e+03	18.500000	6.265000e+02	1.415000e+02	4.000000	0.000000	0.000000	0.945000	48.770000	1.45
50%	5.059000e+03	108.000000	2.815000e+03	1.600000e+03	49.000000	1.000000	22.000000	2.150000	71.320000	3.62
75%	4.046050e+04	734.000000	2.260600e+04	9.149000e+03	419.500000	6.000000	221.000000	3.875000	86.885000	6.44
max	4.290259e+06	148011.000000	1.846641e+06	2.816444e+06	56336.000000	1076.000000	33728.000000	28.560000	100.000000	inf

In [14]:
```python
#Show their columns
data.columns
```

Out[14]:
```
Index(['Country/Region', 'Confirmed', 'Deaths', 'Recovered', 'Active',
       'New_cases', 'New_deaths', 'New_recovered', 'Deaths_/_100_Cases',
       'Recovered_/_100_Cases', 'Deaths_/_100_Recovered',
       'Confirmed_last_week', '1_week_change', '1_week_%_increase',
       'WHO_Region'],
      dtype='object')
```

In [15]:
```python
#Replace column name if it has space
data.rename(columns = {'WHO_Region' :'Regions'},inplace = True)
```

In [16]:
```python
#collect column data and distribute it
data['Regions'].value_counts()
```

Out[16]:
```
Europe                   56
Africa                   48
Americas                 35
Eastern Mediterranean    22
Western Pacific          16
South-East Asia          10
Name: Regions, dtype: int64
```

In [17]:
```python
#Create a table using the name of the columns
pd.crosstab(data.New_deaths, data.Regions,)
```

Out[17]:

Regions	Africa	Americas	Eastern Mediterranean	Europe	South-East Asia	Western Pacific
New_deaths						
0	29	14	6	24	6	12
1	5	3	2	6	0	1
2	5	1	2	3	0	0
3	2	0	2	2	1	0
4	1	1	2	2	0	1
5	2	0	1	3	0	0
6	2	0	0	4	0	1
7	0	0	0	2	0	0
8	1	1	0	0	0	0
9	0	0	1	1	0	0
10	0	0	1	0	0	0
11	0	2	0	1	0	0
13	0	0	0	1	0	1
14	0	0	0	1	0	0
16	0	0	0	1	0	0
17	0	1	0	2	0	0
19	0	0	0	1	0	0
20	0	1	1	0	0	0
24	0	0	0	1	0	0
27	0	1	1	0	0	0
28	0	1	0	0	0	0
37	0	0	0	0	1	0
46	0	0	1	0	0	0
50	0	1	0	0	0	0
57	0	0	0	0	1	0
64	0	1	0	0	0	0
75	0	1	0	0	0	0
85	0	0	0	1	0	0
96	0	0	1	0	0	0
120	0	1	0	0	0	0
212	0	0	1	0	0	0
298	1	0	0	0	0	0
342	0	1	0	0	0	0
508	0	1	0	0	0	0
575	0	1	0	0	0	0
614	0	1	0	0	0	0
637	0	0	0	0	1	0
1076	0	1	0	0	0	0

In [18]:
```python
#Change column name
data.rename(columns = {'Country/Region' :'Country_Region'},inplace = True)
```

In [19]:
```python
#Counting the data in a column
data['Country_Region'].value_counts()
```

Out[19]:
```
Cambodia       1
Bulgaria       1
Belarus        1
Ethiopia       1
Maldives       1
              ..
Burundi        1
Zimbabwe       1
Afghanistan    1
Gambia         1
Liberia        1
Name: Country_Region, Length: 187, dtype: int64
```

In [20]:
```python
#Simple counting of all numerical data
group_data = data.groupby("Regions")
mean_data = group_data.mean()
```

In [21]:
```python
mean_data = mean_data.reset_index()
```

In [22]:
```python
print(mean_data)
```

```
                 Regions      Confirmed       Deaths      Recovered  \
0                 Africa   15066.812500   254.645833    9180.104167   
1               Americas  252551.028571  9792.342857  127674.742857   
2  Eastern Mediterranean   67761.090909  1742.681818   54609.090909   
3                 Europe   58920.053571  3770.428571   35602.196429   
4        South-East Asia  183529.700000  4134.900000  115693.300000   
5        Western Pacific   18276.750000   515.562500   12923.125000   

          Active     New_cases  New_deaths  New_recovered  Deaths_/_100_Cases  \
0    5632.062500   253.666667    7.354167     303.395833            2.306458   
1  115083.942857  3700.885714  101.571429    2707.885714            3.052571   
2   11409.318182   564.090909   20.227273     674.681818            3.563182   
3   19547.428571   398.107143    5.428571     209.500000            4.198393   
4   63701.500000  4899.300000   73.400000    3758.200000            1.296000   
5    4838.062500   205.562500    1.500000      70.437500            1.290000   

   Recovered_/_100_Cases  Deaths_/_100_Recovered  Confirmed_last_week  \
0              57.014792                     inf         12669.166667   
1              62.291429                     inf        223291.371429   
2              66.593182                     inf         63583.954545   
3              68.635000                     inf         56193.142857   
4              66.704000                  1.9560        147828.300000   
5              76.805000                  1.7875         16647.437500   

   1_week_change  1_week_%_increase  
0    2397.645833          18.086458  
1   29259.657143          16.331143  
2    4177.136364          10.482273  
3    2726.910714           7.769643  
4   35701.400000           8.513000  
5    1629.312500          22.111250  
```

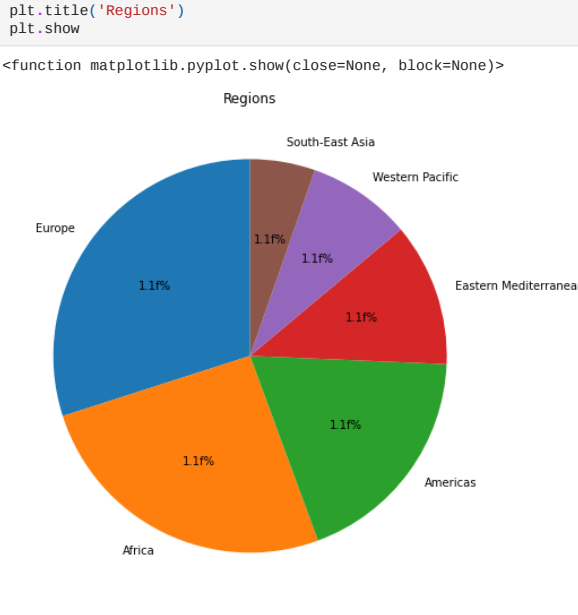
In [23]:
```python
#Average data column
data['Deaths'].mean()
```

Out[23]: 3497.51871657754

In [24]:
```python
#create a figure
labels = data['Regions'].value_counts().index
values = data['Regions'].value_counts().values
```

In [25]:
```python
plt.figure(figsize=(15,8))
plt.pie(values, labels=labels, autopct = '1.1f%%',startangle = 90)
plt.title('Regions')
plt.show
```

Out[25]: <function matplotlib.pyplot.show(close=None, block=None)>



Predicting based on ther continents

In [26]:
```python
data4['Continent'].value_counts()
```

Out[26]:
```
Africa               57
Asia                 48
Europe               48
North America        35
South America        14
Australia/Oceania     6
0                     1
Name: Continent, dtype: int64
```

In [27]:
```python
data4[data4.Continent == "Asia"]
```

Out[27]:

	Country/Region	Continent	Population	TotalCases	NewCases	TotalDeaths	NewDeaths	TotalRecovered	NewRecovered	ActiveCases	Serious,Critical	Tot_Cases/1M_pop
2	India	Asia	1.381345e+09	2025409	0.0	41638.0	0.0	1377384.0	0.0	606387.0	8944.0	1466.0
10	Iran	Asia	8.409762e+07	320117	0.0	17976.0	0.0	277463.0	0.0	24678.0	4156.0	3806.0
12	Saudi Arabia	Asia	3.486592e+07	284226	0.0	3055.0	0.0	247089.0	0.0	34082.0	1915.0	8152.0
13	Pakistan	Asia	2.212959e+08	281863	0.0	6035.0	0.0	256058.0	0.0	19770.0	809.0	1274.0
14	Bangladesh	Asia	1.648514e+08	249651	0.0	3306.0	0.0	143824.0	0.0	102521.0	0.0	1514.0
16	Turkey	Asia	8.442833e+07	237265	0.0	5798.0	0.0	220546.0	0.0	10921.0	580.0	2810.0
20	Iraq	Asia	4.030602e+07	140603	0.0	5161.0	0.0	101025.0	0.0	34417.0	517.0	3488.0
21	Philippines	Asia	1.097227e+08	119460	0.0	2150.0	0.0	66837.0	0.0	50473.0	239.0	1089.0
22	Indonesia	Asia	2.738084e+08	118753	0.0	5521.0	0.0	75645.0	0.0	37587.0	0.0	434.0
24	Qatar	Asia	2.807805e+06	112092	0.0	178.0	0.0	108831.0	0.0	3083.0	77.0	39922.0
25	Kazakhstan	Asia	1.879867e+07	95942	0.0	1058.0	0.0	68871.0	0.0	26013.0	221.0	5104.0
30	Oman	Asia	5.118446e+06	80713	0.0	492.0	0.0	70910.0	0.0	9311.0	177.0	15769.0
31	Israel	Asia	9.197590e+06	79559	0.0	576.0	0.0	53427.0	0.0	25556.0	358.0	8650.0
36	Kuwait	Asia	4.276658e+06	70045	0.0	469.0	0.0	61610.0	0.0	7966.0	127.0	16378.0
38	UAE	Asia	9.902079e+06	61845	0.0	354.0	0.0	55739.0	0.0	5752.0	0.0	6246.0
41	Singapore	Asia	5.854932e+06	54555	0.0	27.0	0.0	48031.0	0.0	6497.0	0.0	9318.0
47	Bahrain	Asia	1.706669e+06	42889	0.0	156.0	0.0	39945.0	0.0	2788.0	41.0	25130.0
48	Japan	Asia	1.264359e+08	42263	0.0	1026.0	0.0	28877.0	0.0	12360.0	115.0	334.0
49	Armenia	Asia	2.963811e+06	39819	0.0	772.0	0.0	31556.0	0.0	7491.0	0.0	13435.0
51	Kyrgyzstan	Asia	6.534479e+06	38659	0.0	1447.0	0.0	30099.0	0.0	7113.0	24.0	5916.0
52	Afghanistan	Asia	3.900945e+07	36896	0.0	1298.0	0.0	25840.0	0.0	9758.0	31.0	946.0
55	Azerbaijan	Asia	1.014824e+07	33247	0.0	479.0	0.0	29275.0	0.0	3493.0	66.0	3276.0
57	Uzbekistan	Asia	3.351603e+07	28315	0.0	175.0	0.0	19291.0	0.0	8849.0	228.0	845.0
63	Nepal	Asia	2.918649e+07	21750	0.0	65.0	0.0	15389.0	0.0	6296.0	0.0	745.0
72	S. Korea	Asia	5.127373e+07	14519	20.0	303.0	1.0	13543.0	42.0	673.0	18.0	283.0
74	Palestine	Asia	5.112340e+06	13398	0.0	92.0	0.0	6907.0	0.0	6399.0	0.0	2621.0
83	Malaysia	Asia	3.240637e+07	9038	0.0	125.0	0.0	8713.0	0.0	200.0	2.0	279.0
86	Tajikistan	Asia	9.557468e+06	7665	0.0	62.0	0.0	6443.0	0.0	1160.0	0.0	802.0
95	Lebanon	Asia	6.822220e+06	5672	0.0	70.0	0.0	1974.0	0.0	3628.0	46.0	831.0
101	Maldives	Asia	5.414480e+05	4680	0.0	19.0	0.0	2725.0	0.0	1936.0	12.0	8643.0
107	Hong Kong	Asia	7.503041e+06	3850	0.0	46.0	0.0	2458.0	0.0	1346.0	39.0	513.0
110	Thailand	Asia	6.981789e+07	3330	0.0	58.0	0.0	3148.0	0.0	124.0	1.0	48.0
114	Sri Lanka	Asia	2.142236e+07	2839	0.0	11.0	0.0	2541.0	0.0	287.0	1.0	133.0
131	Yemen	Asia	2.988690e+07	1768	0.0	508.0	0.0	898.0	0.0	362.0	0.0	59.0
137	Jordan	Asia	1.021314e+07	1232	0.0	11.0	0.0	1171.0	0.0	50.0	3.0	121.0
140	Cyprus	Asia	1.208238e+06	1208	0.0	19.0	0.0	856.0	0.0	333.0	0.0	1000.0
141	Georgia	Asia	3.988368e+06	1206	0.0	17.0	0.0	987.0	0.0	202.0	0.0	302.0
145	Syria	Asia	1.753960e+07	999	0.0	48.0	0.0	311.0	0.0	640.0	0.0	57.0
154	Vietnam	Asia	9.742547e+07	747	0.0	10.0	0.0	392.0	0.0	345.0	0.0	8.0
162	Taiwan	Asia	2.382120e+07	477	0.0	7.0	0.0	443.0	0.0	27.0	0.0	20.0
165	Myanmar	Asia	5.444639e+07	357	0.0	6.0	0.0	308.0	0.0	43.0	0.0	7.0
168	Mongolia	Asia	3.283344e+06	293	0.0	0.0	0.0	260.0	0.0	33.0	1.0	89.0
174	Cambodia	Asia	1.674138e+07	243	0.0	0.0	0.0	210.0	0.0	33.0	1.0	15.0
181	Brunei	Asia	4.378930e+05	141	0.0	3.0	0.0	138.0	0.0	0.0	0.0	322.0
186	Bhutan	Asia	7.724430e+05	105	0.0	0.0	0.0	93.0	0.0	12.0	0.0	136.0
193	Macao	Asia	6.501930e+05	46	0.0	0.0	0.0	46.0	0.0	0.0	0.0	71.0
197	Timor-Leste	Asia	1.320812e+06	25	0.0	0.0	0.0	24.0	0.0	1.0	0.0	19.0
200	Laos	Asia	7.285750e+06	20	0.0	0.0	0.0	19.0	0.0	1.0	0.0	3.0

i would like choose some countries to do a feasibility study

In [28]:
```python
#Access for any row
our_contry = data4.iloc[[12,20,24,30,36,38,47,131,137,145]]
our_contry
```

Out[28]:

	Country/Region	Continent	Population	TotalCases	NewCases	TotalDeaths	NewDeaths	TotalRecovered	NewRecovered	ActiveCases	Serious,Critical	Tot_Cases/1M_pop
12	Saudi Arabia	Asia	34865919.0	284226	0.0	3055.0	0.0	247089.0	0.0	34082.0	1915.0	8152.0
20	Iraq	Asia	40306025.0	140603	0.0	5161.0	0.0	101025.0	0.0	34417.0	517.0	3488.0
24	Qatar	Asia	2807805.0	112092	0.0	178.0	0.0	108831.0	0.0	3083.0	77.0	39922.0
30	Oman	Asia	5118446.0	80713	0.0	492.0	0.0	70910.0	0.0	9311.0	177.0	15769.0
36	Kuwait	Asia	4276658.0	70045	0.0	469.0	0.0	61610.0	0.0	7966.0	127.0	16378.0
38	UAE	Asia	9902079.0	61845	0.0	354.0	0.0	55739.0	0.0	5752.0	0.0	6246.0
47	Bahrain	Asia	1706669.0	42889	0.0	156.0	0.0	39945.0	0.0	2788.0	41.0	25130.0
131	Yemen	Asia	29886897.0	1768	0.0	508.0	0.0	898.0	0.0	362.0	0.0	59.0
137	Jordan	Asia	10213138.0	1232	0.0	11.0	0.0	1171.0	0.0	50.0	3.0	121.0
145	Syria	Asia	17539600.0	999	0.0	48.0	0.0	311.0	0.0	640.0	0.0	57.0

Display Each Country and their information

Saudi Arabia

In [29]:
```python
data4.xs(12)
```

Out[29]:
```
Country/Region               Saudi Arabia
Continent                            Asia
Population                    3.48659e+07
TotalCases                         284226
NewCases                                0
TotalDeaths                          3055
NewDeaths                               0
TotalRecovered                     247089
NewRecovered                            0
ActiveCases                         34082
Serious,Critical                     1915
Tot_Cases/1M_pop                     8152
Deaths/1M_pop                          88
TotalTests                    3.63570e+06
Tests/1M_pop                       104277
WHO_Region          EasternMediterranean
Name: 12, dtype: object
```

Iraq

In [30]:
```python
data4.xs(20)
```

Out[30]:
```
Country/Region                       Iraq
Continent                            Asia
Population                    4.0306e+07
TotalCases                         140603
NewCases                                0
TotalDeaths                          5161
NewDeaths                               0
TotalRecovered                     101025
NewRecovered                            0
ActiveCases                         34417
Serious,Critical                      517
Tot_Cases/1M_pop                     3488
Deaths/1M_pop                         128
TotalTests                    1.09274e+06
Tests/1M_pop                        27111
WHO_Region          EasternMediterranean
Name: 20, dtype: object
```

Qatar

In [31]:
```python
data4.xs(24)
```

Out[31]:
```
Country/Region                      Qatar
Continent                            Asia
Population                    2.80780e+06
TotalCases                         112092
NewCases                                0
TotalDeaths                           178
NewDeaths                               0
TotalRecovered                     108831
NewRecovered                            0
ActiveCases                          3083
Serious,Critical                       77
Tot_Cases/1M_pop                    39922
Deaths/1M_pop                          63
TotalTests                         511000
Tests/1M_pop                       181993
WHO_Region          EasternMediterranean
Name: 24, dtype: object
```

Oman

In [32]:
```python
data4.xs(30)
```

Out[32]:
```
Country/Region                       Oman
Continent                            Asia
Population                    5.11845e+06
TotalCases                          80713
NewCases                                0
TotalDeaths                           492
NewDeaths                               0
TotalRecovered                      70910
NewRecovered                            0
ActiveCases                          9311
Serious,Critical                      177
Tot_Cases/1M_pop                    15769
Deaths/1M_pop                          96
TotalTests                         309212
Tests/1M_pop                        60411
WHO_Region          EasternMediterranean
Name: 30, dtype: object
```

Kuwait

In [33]:
```python
data4.xs(36)
```

Out[33]:
```
Country/Region                     Kuwait
Continent                            Asia
Population                    4.27666e+06
TotalCases                          70045
NewCases                                0
TotalDeaths                           469
NewDeaths                               0
TotalRecovered                      61610
NewRecovered                            0
ActiveCases                          7966
Serious,Critical                      127
Tot_Cases/1M_pop                    16378
Deaths/1M_pop                         110
TotalTests                         522200
Tests/1M_pop                       122105
WHO_Region          EasternMediterranean
Name: 36, dtype: object
```

UAE

In [34]:
```python
data4.xs(38)
```

Out[34]:
```
Country/Region                        UAE
Continent                            Asia
Population                    9.90208e+06
TotalCases                          61845
NewCases                                0
TotalDeaths                           354
NewDeaths                               0
TotalRecovered                      55739
NewRecovered                            0
ActiveCases                          5752
Serious,Critical                        0
Tot_Cases/1M_pop                     6246
Deaths/1M_pop                          36
TotalTests                    5.26266e+06
Tests/1M_pop                       531470
WHO_Region          EasternMediterranean
Name: 38, dtype: object
```

Bahrain

In [35]:
```python
data4.xs(47)
```

Out[35]:
```
Country/Region                    Bahrain
Continent                            Asia
Population                    1.70667e+06
TotalCases                          42889
NewCases                                0
TotalDeaths                           156
NewDeaths                               0
TotalRecovered                      39945
NewRecovered                            0
ActiveCases                          2788
Serious,Critical                       41
Tot_Cases/1M_pop                    25130
Deaths/1M_pop                          91
TotalTests                         876700
Tests/1M_pop                       513691
WHO_Region          EasternMediterranean
Name: 47, dtype: object
```

Yemen

In [36]:
```python
data4.xs(131)
```

Out[36]:
```
Country/Region                      Yemen
Continent                            Asia
Population                    2.98869e+07
TotalCases                           1768
NewCases                                0
TotalDeaths                           508
NewDeaths                               0
TotalRecovered                        898
NewRecovered                            0
ActiveCases                           362
Serious,Critical                        0
Tot_Cases/1M_pop                       59
Deaths/1M_pop                          17
TotalTests                            120
Tests/1M_pop                            4
WHO_Region          EasternMediterranean
Name: 131, dtype: object
```

Jordan

In [37]:
```python
data4.xs(137)
```

Out[37]:
```
Country/Region                     Jordan
Continent                            Asia
Population                    1.02131e+07
TotalCases                           1232
NewCases                                0
TotalDeaths                            11
NewDeaths                               0
TotalRecovered                       1171
NewRecovered                            0
ActiveCases                            50
Serious,Critical                        3
Tot_Cases/1M_pop                      121
Deaths/1M_pop                           1
TotalTests                         628745
Tests/1M_pop                        61562
WHO_Region          EasternMediterranean
Name: 137, dtype: object
```

Syria

In [38]:
```python
data4.xs(145)
```

Out[38]:
```
Country/Region                      Syria
Continent                            Asia
Population                    1.75396e+07
TotalCases                            999
NewCases                                0
TotalDeaths                            48
NewDeaths                               0
TotalRecovered                        311
NewRecovered                            0
ActiveCases                           640
Serious,Critical                        0
Tot_Cases/1M_pop                       57
Deaths/1M_pop                           3
TotalTests                              0
Tests/1M_pop                            0
WHO_Region          EasternMediterranean
Name: 145, dtype: object
```

Let's do simple Comparison

In [39]:
```python
s1 = pd.Series([1,2])
s2 = pd.Series(["Saudi Arabia", "Bahrain"])
data_four1 = pd.DataFrame([s1,s2])
data_four1
dframe = pd.DataFrame([[1,2],["Saudi Arabia", "Bahrain"]],
        index=["r1", "r2"],
        columns=["c1", "c2"])
dframe
dframe = pd.DataFrame({
        "Population1": [34865919.0, "Saudi Arabia"],
        "Population2": [ 1.70667e+06, "Bahrain"]})
dframe
```

Out[39]:

	Population1	Population2
0	3.48659e+07	1.70667e+06
1	Saudi Arabia	Bahrain

We can say the Saudi Arabia has Higher population than other Countries

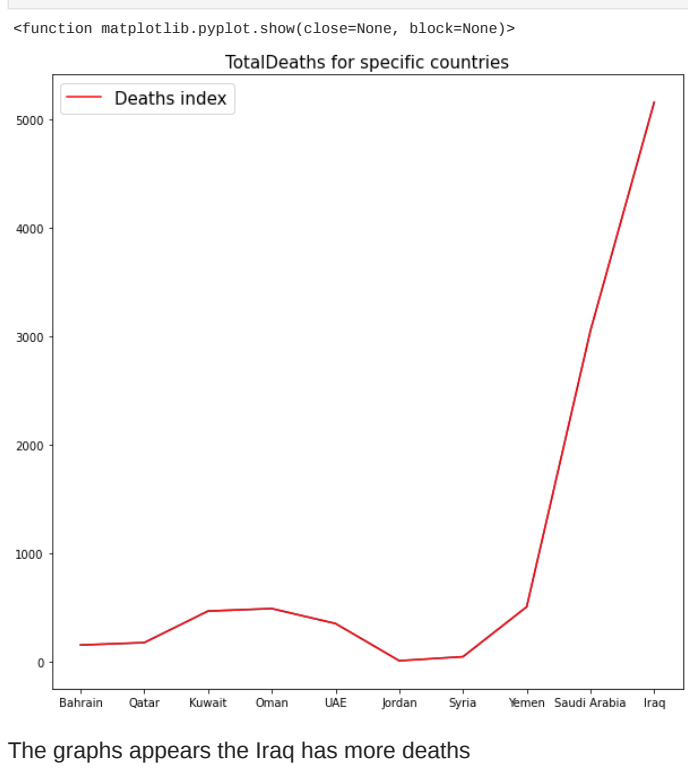
On other hand Bahrain is the lowest

'Now we have three Comparisons'

- first one

In [40]:
```python
Country_a = (['Bahrain','Qatar','Kuwait','Oman','UAE','Jordan','Syria','Yemen','Saudi Arabia','Iraq'])
Population_b = ([156,178,469,492,354,11,48,508,3055,5161])
plt.figure(figsize=(10,10))
plt.plot(Country_a,Population_b)
plt.plot(Population_b,color='red',label='Deaths index')
plt.title("TotalDeaths for specific countries",fontsize = 15)
plt.legend(loc='upper left',fontsize=15)
plt.show
```

Out[40]: <function matplotlib.pyplot.show(close=None, block=None)>



The graphs appears the Iraq has more deaths

but also it's close to Saudi Arabia

In [42]:
```python
s1 = pd.Series([1,2])
s2 = pd.Series(["Saudi Arabia", "Iraq"])
data_four = pd.DataFrame([s1,s2])
data_four
dframe = pd.DataFrame([[1,2],["Saudi Arabia", "Iraq"]],
        index=["r1", "r2"],
        columns=["c1", "c2"])
dframe
dframe = pd.DataFrame({
        "Population1": [34865919.0, "Saudi Arabia"],
        "Population2": [1706669.0, "Iraq"]})
dframe
```

Out[42]:

	Population1	Population2
0	3.48659e+07	1.70667e+06
1	Saudi Arabia	Iraq

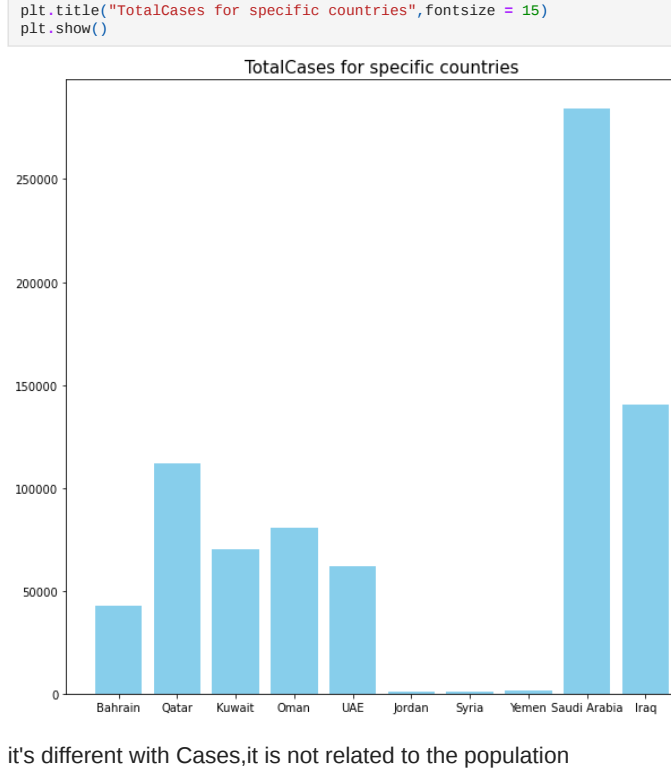
From the comparison above, we can predict if the population higher the deaths will rise (positive relationship)

- second one

In [43]:
```python
category = ['Bahrain','Qatar','Kuwait','Oman','UAE','Jordan','Syria','Yemen','Saudi Arabia','Iraq']
TotalCases = np.array([42889,112092,70045,80713,61845,1232,999,1768,284226,140603])

#plot

plt.figure(figsize=(10,10))
plt.bar(range(0,10), TotalCases,color='skyblue')
plt.xticks(range(0,10),category)
plt.title("TotalCases for specific countries",fontsize = 15)
plt.show()
```

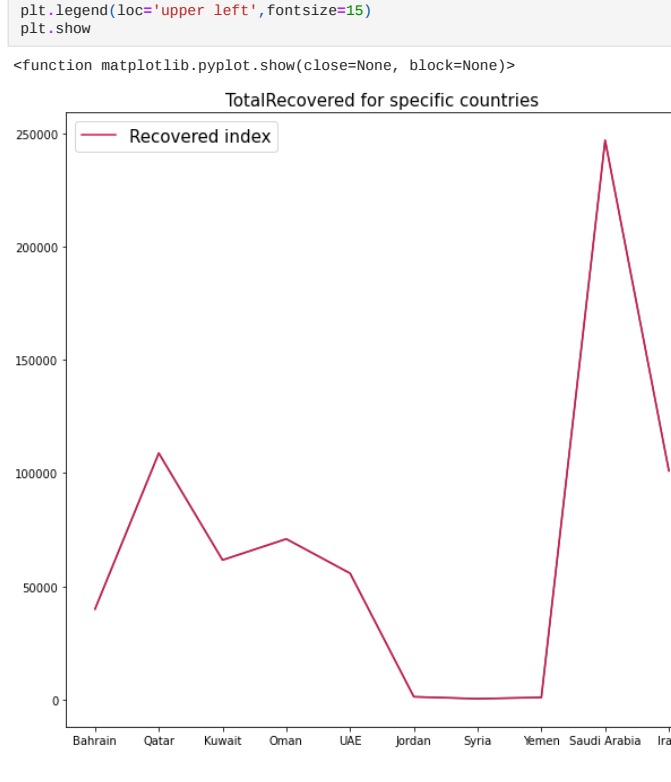


it's different with Cases,it is not related to the population

- third one

In [44]:
```python
Country_a = (['Bahrain','Qatar','Kuwait','Oman','UAE','Jordan','Syria','Yemen','Saudi Arabia','Iraq'])
TotalRecovered= ([39945,108831,61610,70910,55739,1171,311,898,247089,101025])
plt.figure(figsize=(10,10))
plt.plot(Country_a,TotalRecovered)
plt.plot(TotalRecovered,color='crimson',label='Recovered index')
plt.title("TotalRecovered for specific countries",fontsize = 15)
plt.legend(loc='upper left',fontsize=15)
plt.show
```

Out[44]: <function matplotlib.pyplot.show(close=None, block=None)>



Unfortunately, i think that give us an impression of the power of medicine for each country

The prove, bahrain the lowest population but has more recovered. However, syria and jordan must looking for solution, such as build hospital

We will Focus in EasternMediterranean

Let's see how many countries from EasternMediterranean

and collected them in General

i'll show it below

In [45]:
```python
MiddleEast = []
for i in data4.WHO_Region:
    if i == 'EasternMediterranean':
        MiddleEast.append(True)
    else:
        MiddleEast.append(False)
```

In [46]:
```python
is_long = pd.Series(MiddleEast)
```

In [47]:
```python
data4[is_long]
```

Out[47]:

	Country/Region	Continent	Population	TotalCases	NewCases	TotalDeaths	NewDeaths	TotalRecovered	NewRecovered	ActiveCases	Serious,Critical	Tot_Cases/1M_pop
10	Iran	Asia	84097623.0	320117	0.0	17976.0	0.0	277463.0	0.0	24678.0	4156.0	3806.0
12	Saudi Arabia	Asia	34865919.0	284226	0.0	3055.0	0.0	247089.0	0.0	34082.0	1915.0	8152.0
13	Pakistan	Asia	221295851.0	281863	0.0	6035.0	0.0	256058.0	0.0	19770.0	809.0	1274.0
20	Iraq	Asia	40306025.0	140603	0.0	5161.0	0.0	101025.0	0.0	34417.0	517.0	3488.0
24	Qatar	Asia	2807805.0	112092	0.0	178.0	0.0	108831.0	0.0	3083.0	77.0	39922.0
26	Egypt	Africa	102516525.0	95006	0.0	4951.0	0.0	48898.0	0.0	41157.0	41.0	927.0
30	Oman	Asia	5118446.0	80713	0.0	492.0	0.0	70910.0	0.0	9311.0	177.0	15769.0
36	Kuwait	Asia	4276658.0	70045	0.0	469.0	0.0	61610.0	0.0	7966.0	127.0	16378.0
38	UAE	Asia	9902079.0	61845	0.0	354.0	0.0	55739.0	0.0	5752.0	0.0	6246.0
47	Bahrain	Asia	1706669.0	42889	0.0	156.0	0.0	39945.0	0.0	2788.0	41.0	25130.0
52	Afghanistan	Asia	39009447.0	36896	0.0	1298.0	0.0	25840.0	0.0	9758.0	31.0	946.0
56	Morocco	Africa	36953359.0	29644	0.0	449.0	0.0	20553.0	0.0	8642.0	31.0	802.0
74	Palestine	Asia	5112340.0	13398	0.0	92.0	0.0	6907.0	0.0	6399.0	0.0	2621.0
78	Sudan	Africa	43943536.0	11780	0.0	763.0	0.0	6194.0	0.0	4823.0	0.0	268.0
95	Lebanon	Asia	6822220.0	5672	0.0	70.0	0.0	1974.0	0.0	3628.0	46.0	831.0
97	Djibouti	Africa	989387.0	5330	0.0	59.0	0.0	5057.0	0.0	214.0	0.0	5387.0
99	Libya	Africa	6880353.0	4879	0.0	107.0	0.0	652.0	0.0	4120.0	0.0	709.0
111	Somalia	Africa	15933012.0	3227	0.0	93.0	0.0	1728.0	0.0	1406.0	2.0	203.0
131	Yemen	Asia	29886897.0	1768	0.0	508.0	0.0	898.0	0.0	362.0	0.0	59.0
132	Tunisia	Africa	11830801.0	1642	0.0	51.0	0.0	1241.0	0.0	350.0	9.0	139.0
137	Jordan	Asia	10213138.0	1232	0.0	11.0	0.0	1171.0	0.0	50.0	3.0	121.0
145	Syria	Asia	17539600.0	999	0.0	48.0	0.0	311.0	0.0	640.0	0.0	57.0

Mission Done

In [48]:
```python
#Assign a Varibale
x= data4[is_long]
```

information in EasternMediterranean

Total population

In [49]:
```python
s0 = x['Population'].sum()
data_table = pd.DataFrame([s0])
dframe = pd.DataFrame(s0,columns= ["Total population"],index=[""])
dframe
```

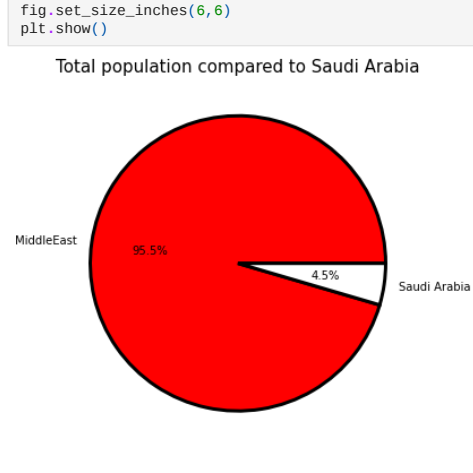
Out[49]:

	Total population
	732007690.0

In [50]:
```python
Populationx = ['MiddleEast','Saudi Arabia']

P =[732007690.0,34865919.0]
```

In [51]:
```python
fig,axs=plt.subplots()
axs.pie(P,labels=Populationx,autopct = '%1.1f%%',wedgeprops = { 'linewidth' : 3, 'edgecolor' : 'black' }
,colors=['r','white'])
axs.set_title('Total population compared to Saudi Arabia',fontsize=15)
fig.set_size_inches(6,6)
plt.show()
```



TotalCases

In [52]:
```python
s1 = x['TotalCases'].sum()
data_table = pd.DataFrame([s1])
dframe = pd.DataFrame(s1,columns= ["TotalCases"],index=[""])
dframe
```

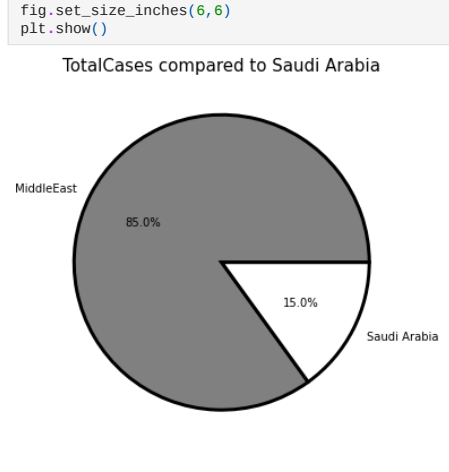
Out[52]:

	TotalCases
	1605866

In [53]:
```python
Casesx = ['MiddleEast','Saudi Arabia']

C=[1605866,284226]
```

In [54]:
```python
fig,axs=plt.subplots()
axs.pie(C,labels=Casesx,autopct = '%1.1f%%',wedgeprops = { 'linewidth' : 3, 'edgecolor' : 'black' }
,colors=['grey','white'])
axs.set_title('TotalCases compared to Saudi Arabia',fontsize=15)
fig.set_size_inches(6,6)
plt.show()
```



TotalDeaths

In [55]:
```python
s2 = x['TotalDeaths'].sum()
data_table = pd.DataFrame([s2])
dframe = pd.DataFrame(s2,columns= ["TotalDeaths"],index=[""])
dframe
```

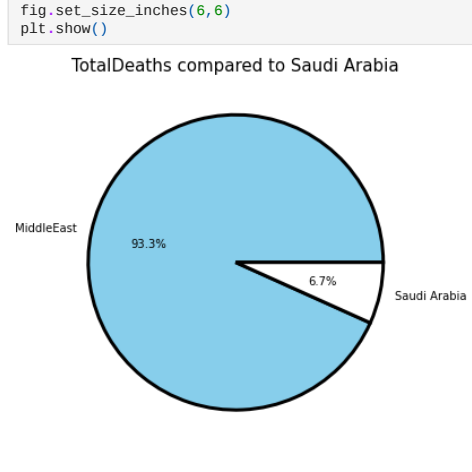
Out[55]:

	TotalDeaths
	42376.0

In [56]:
```python
Deathx = ['MiddleEast','Saudi Arabia']

D=[42376.0,3055.0]
```

In [57]:
```python
fig,axs=plt.subplots()
axs.pie(D,labels=Deathx,autopct = '%1.1f%%',wedgeprops = { 'linewidth' : 3, 'edgecolor' : 'black' }
,colors=['skyblue','white'])
axs.set_title('TotalDeaths compared to Saudi Arabia',fontsize=15)
fig.set_size_inches(6,6)
plt.show()
```



TotalRecovered

In [58]:
```python
s3 = x['TotalRecovered'].sum()
data_table = pd.DataFrame([s3])
dframe = pd.DataFrame(s3,columns= ["TotalRecovered"],index=[""])
dframe
```

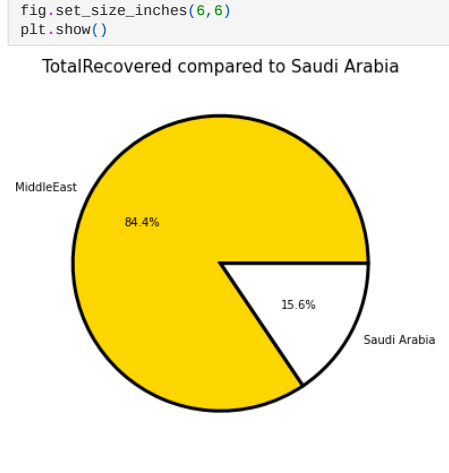
Out[58]:

	TotalRecovered
	1340094.0

In [59]:
```python
Recoveredx = ['MiddleEast','Saudi Arabia']

R=[1340094.0,247089.0]
```

In [60]:
```python
fig,axs=plt.subplots()
axs.pie(R,labels=Recoveredx,autopct = '%1.1f%%',wedgeprops = { 'linewidth' : 3, 'edgecolor' : 'black' }
,colors=['gold','white'])
axs.set_title('TotalRecovered compared to Saudi Arabia',fontsize=15)
fig.set_size_inches(6,6)
plt.show()
```



Total ActiveCases

In [61]:
```python
s4 = x['ActiveCases'].sum()
data_table = pd.DataFrame([s4])
dframe = pd.DataFrame(s4,columns= ["Total ActiveCases"],index=[""])
dframe
```

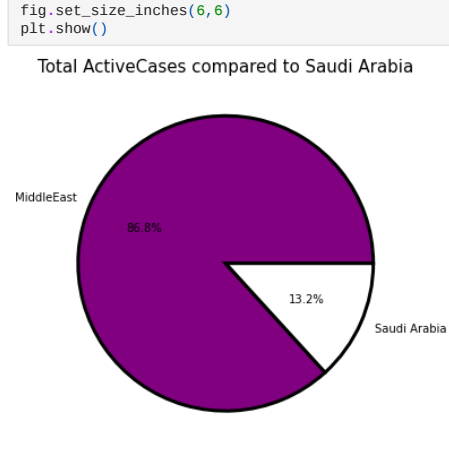
Out[61]:

	Total ActiveCases
	223396.0

In [62]:
```python
ActiveCasesx = ['MiddleEast','Saudi Arabia']

A =[223396.0,34082.0]
```

In [63]:
```python
fig,axs=plt.subplots()
axs.pie(A,labels=ActiveCasesx,autopct = '%1.1f%%',wedgeprops = { 'linewidth' : 3, 'edgecolor' : 'black' }
,colors=['purple','white'])
axs.set_title('Total ActiveCases compared to Saudi Arabia',fontsize=15)
fig.set_size_inches(6,6)
plt.show()
```



Total Serious,Critical

In [64]:
```python
s5 = x['Serious,Critical'].sum()
data_table = pd.DataFrame([s5])
dframe = pd.DataFrame(s5,columns= ["Total Serious,Critical"],index=[""])
dframe
```

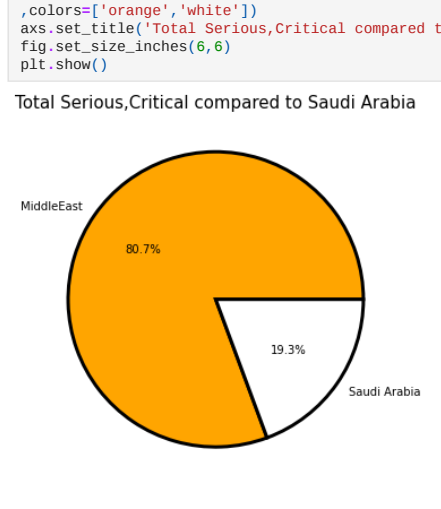
Out[64]:

	Total Serious,Critical
	7982.0

In [65]:
```python
Seriousx = ['MiddleEast','Saudi Arabia']

S =[7982.0,1915.0]
```

In [66]:
```python
fig,axs=plt.subplots()
axs.pie(S,labels=Seriousx,autopct = '%1.1f%%',wedgeprops = { 'linewidth' : 3, 'edgecolor' : 'black' }
,colors=['orange','white'])
axs.set_title('Total Serious,Critical compared to Saudi Arabia',fontsize=15)
fig.set_size_inches(6,6)
plt.show()
```



TotalTests

In [67]:
```python
s6 = x['TotalTests'].sum()
data_table = pd.DataFrame([s6])
dframe = pd.DataFrame(s6,columns= ["TotalTests"],index=[""])
dframe
```

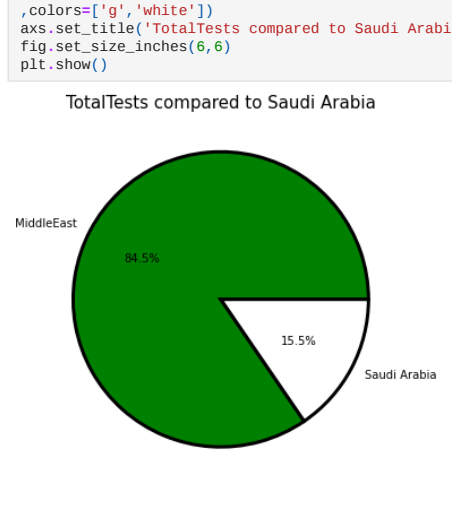
Out[67]:

	TotalTests
	19885783.0

In [68]:
```python
Testsx = ['MiddleEast','Saudi Arabia']

T =[19885783.0,3635705.0]
```

In [69]:
```python
fig,axs=plt.subplots()
axs.pie(T,labels=Testsx,autopct = '%1.1f%%',wedgeprops = { 'linewidth' : 3, 'edgecolor' : 'black' }
,colors=['g','white'])
axs.set_title('TotalTests compared to Saudi Arabia',fontsize=15)
fig.set_size_inches(6,6)
plt.show()
```



End